

Towards “Nash” Equilibrium for Mixed Agents

Vijay Subramanian

Joint work with Apurv Shukla (University of Michigan), Andy
Zhao (USC) and Rahul Jain (USC)

2025 MURI/TTIC Summer Workshop on Algorithmic Game Theory

July 21, 2025

Agenda

- 1 Nash Equilibria (NE)
 - Basic Tenets of NE
- 2 Generalized Quantal Response Equilibria
 - Quantal Response Equilibria
 - Generalized Quantal Response
 - Generalized Quantal Response Equilibria
 - QGRE Verification
- 3 QGRE Computation with Learning
 - 'RL' Simulator Model
 - Smoothened Frank-Wolfe
 - No Regret
 - Numerical Results
 - Jordan1993's game
- 4 Conclusion

Agenda

- 1 Nash Equilibria (NE)
 - Basic Tenets of NE
- 2 Generalized Quantal Response Equilibria
- 3 GQRE Computation with Learning
- 4 Conclusion

Basic Tenets of NE and Question

- Rationality
- Common Knowledge
- (Infinite) Hierarchy of Beliefs

With mixed agents, some subset or none of these may hold for all

Question: What sort of equilibrium behaviors will result, if any?

Agenda

- 1 Nash Equilibria (NE)
- 2 Generalized Quantal Response Equilibria
 - Quantal Response Equilibria
 - Generalized Quantal Response
 - Generalized Quantal Response Equilibria
 - GQRE Verification
- 3 GQRE Computation with Learning
- 4 Conclusion

Quantal Response Equilibria

- Experimental literature in behavioral economics establishes one class of deviations from Nash equilibrium
- Quantal Response Equilibrium (McKelvey and Palfrey, 1995)
 - Model stochastic choice by considering noisy utilities
 - Equivalently, penalize utility with entropy cost
 - Used to analyze experimental data (Goeree et. al., 2016)

Quantal Response Equilibria

- Can we consider other models of stochastic choice?
 - Related work: Gui et. al. (2025) and Mazumder et. al. (2024)
 - More generally, other cost functions?
- Can this equilibrium be computed efficiently (in a partially decentralized setting)?

For simplicity, focus on 2-player games with mixed strategies

Quantal Response Equilibria: Formal and Standard View

- Finite Normal Form game: $\mathcal{G} \triangleq \{N, A, \{u_i\}_{i \in N}\}$
- (Wlog) Utilities in $[0, 1]$
- Agent i 's response with weighting parameter λ_i , when other agent plays π_{-i} is:

$$\xi_i^{\lambda_i}(\pi_{-i}; p \log(p)) \triangleq \max_{\pi_i \in \Delta(A_i)} \lambda_i u_i(\pi_i, \pi_{-i}) - \pi_i \log \pi_i$$

- Equilibrium: Fixed point of the best-response function
 - Best-response is logit response
 - Either Brouwer or Rosen1965 yield existence

Quantal Response Equilibria: Alternate View

- Agent i 's response with tolerance γ_i , when other agent plays π_{-i} is:

$$\xi_i^{\lambda_i}(\pi_{-i}; p \log(p)) = \min_{\pi_i \in \Delta(A_i)} \pi_i \log \pi_i$$
$$\text{s.t. } u_i(\pi_i, \pi_{-i}) \geq \gamma_i$$

- Equivalent by Lagrangian duality
- $\gamma_i = \max_{\pi_i \in \Delta(A_i)} u_i(\pi_i, \pi_{-i}) \Rightarrow$ best-response
- $\gamma_i = 0 \Rightarrow$ just minimize cost

Interpretation: *Minimize cost of strategy but ensure some minimum utility*

Key idea: γ_i (or λ_i) AND function $p \log(p)$ can be **chosen/set** by agent

This Talk

- A notion of equilibrium: Generalized Quantal Response Equilibrium (GQRE)
- Efficient computation of GQRE in partially decentralized learning

Generalized Quantal Response

Definition (Generalized Quantal Response (GQR))

Given a normal form game $\mathcal{G}$ and a strictly convex function f_i , the (λ_i, f_i) generalized quantal response of agent i , $\xi_i^{\lambda_i}(\pi_{-i}; f_i) \in \Delta(A_i)$, is given by:

$$\xi_i^{\lambda_i}(\pi_{-i}; f_i) \triangleq \arg \max_{\pi_i \in \Delta(A_i)} \lambda_i u_i(\pi_i, \pi_{-i}) - f_i(\pi_i).$$

Examples of f_i :

- ϕ -divergence wrt reference strategy
QR uses KL divergence relative to uniformly random strategy
- Wasserstein distance from reference strategy
- Total variation distance from reference strategy

GQRE: Existence

- GQR exists if f_i is convex, and is unique if f_i is strictly convex
- A pair of policies $\pi^* = (\pi_1^*, \pi_2^*)$ is GQRE if:

$$\forall i \in \{1, 2\}, \pi_i^* \in \xi_i^{\lambda_i}(\pi_{-i}^*; f_i)$$

GQRE is NE of a perturbed game

- Perturbed game $\mathcal{G}^f = \{[N], \{\Delta(A_i)\}_{i \in N}, \{u_i^{f_i}\}_{i \in [N]}\}$ where,
 $u_i^{f_i}(\pi_i, \pi_{-i}) \triangleq \lambda_i u_i(\pi_i, \pi_{-i}) - f_i(\pi_i)$ for any mixed strategy profile $\pi = (\pi_i, \pi_{-i})$

Theorem (GQRE of $\mathcal{G}$ is NE of $\mathcal{G}_f$)

A pair of policies (π_1^, π_2^*) is the GQRE for $\mathcal{G}$ if and only if they are the Nash Equilibria for $\mathcal{G}_f$.*

- GQRE exists for f_i convex by Rosen1965
- Algorithms for NE computation of perturbed game can be used for GQRE computation

GQRE verification is tractable

- Algorithmic verification mechanism to check whether a given strategy profile (π_i, π_{-i}) is a GQRE can be done in time linear in the action space and the number of players.

Lemma (Verification of GQRE)

A strategy profile π^ is a GQRE if and only if*

$$\max_{a_i \in A_i} \langle a_i - \pi_i^*, \nabla_{\pi_i} u_i^{f_i}(\pi^*) \rangle \leq 0, \quad \forall i \in [N].$$

- Proof: Variational inequality characterization plus polyhedral nature of set of mixed strategies

Characterization core to gradient-based computational schemes

Agenda

- 1 Nash Equilibria (NE)
- 2 Generalized Quantal Response Equilibria
- 3 GQRE Computation with Learning
 - 'RL' Simulator Model
 - Smoothened Frank-Wolfe
 - No Regret
 - Numerical Results
 - Jordan1993's game
- 4 Conclusion

Issues with the state of the art

Learning-based setup: Noisy gradients provided to agents

Standard noisy oracle model in the literature

- Unbiased gradient
- Conditional variance bounded for all strategy profiles

Issues: Doesn't capture troubles of rarely using some action(s)

Computational schemes in the literature centralized

Proximal gradient, Extra gradient, Dual averaging, etc.

Projection involved is primary computational bottleneck

Question: Can we find fully or partially decentralized schemes with projection overhead being low?

A realistic 'RL' oracle model

Simulator oracle for game play:

- Agent's input strategies to oracle that knows the game
- Oracle simulates $\mathcal{G}^f$, $M \geq 1$ times with given strategy profile $\{\mathbf{A}(m) = (A_1(m), \dots, A_N(m))\}_{m=1}^M$ i.i.d. $\prod_{i \in [N]} \pi_i$
- Each agents receives for every action empirical count and cumulative payoff over entire play

$$U_i(a_i) = \sum_{m=1}^M u_i(\mathbf{A}(m)) 1_{\{A_i(m)=a_i\}}$$

$$\mathbb{E}[U_i(a_i)] = M\pi_i(a_i)u_i(a_i, \pi_{-i}).$$

Unbiased gradient estimate when $\pi_i(a_i) > 0$:

$$\hat{F}_i(a_i) = \frac{\lambda_i}{\pi_i(a_i)} \frac{U(a_i)}{M}, \quad \text{Var}(\hat{F}_i(a_i)) = \frac{\lambda_i^2 \widetilde{\text{Var}}(a_i, \pi_i, \pi_{-i}; u_i)}{M\pi_i(a_i)}, \quad \forall a_i \in A_i.$$

GQRE Computation – Smoothened Frank-Wolfe

- Frank-Wolfe algorithm: projection-free method but set-valued
- Smoothened Frank-Wolfe: Choose for all i

$$s_i^*(\pi) \triangleq \arg \max_{s \in \Delta(A_i)} \langle s, \nabla_{\pi_i} u_i^{f_i}(\pi) \rangle - \eta \text{KL}(s, \pi_i), \quad \forall i \in [N],$$

- Explicitly given by logit response
- Project $s_i^*(\pi)$ to ϵ -exploration simplex $\Delta(A_i; \epsilon)$, $\epsilon \in (0, \frac{1}{|A_i|}]$
 - Avoids variance blow up
From simulator & gradient of KL divergence
 - Easy to compute via linear search
- Each agent i takes a small step towards projection from π_i

Assumptions for analysis of algorithm

We assume the following for the perturbed utilities:

- Lipschitz continuous gradients of the perturbed utilities
- Lipschitz-continuous Jacobian matrix H

$H(\pi) + H^T(\pi)$ is negative definite $\forall \pi \in \Delta(A_1) \times \Delta(A_2)$

Implies uniqueness of GQRE by Rosen1965!

GQRE Computation—No Regret

- New smooth gap function: ≥ 0 & 0 only for GQRE

$$V(\pi) = \sum_i \max_{s \in \Delta(A_i)} \langle s - \pi_i, \nabla_{\pi_i} u_i^{f_i}(\pi) \rangle - \eta \text{KL}(s, \pi_i)$$

- This is a Lyapunov function for smoothened FW

Theorem (No-regret guarantee)

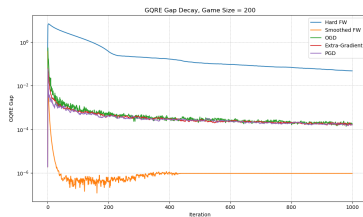
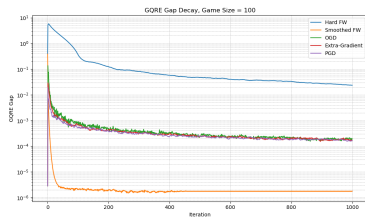
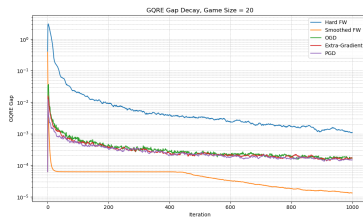
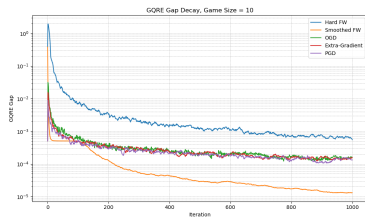
A time-step t step-size $\gamma_t = \frac{1}{t+1}$, $\epsilon_t = \frac{1}{(t+1) \max_i |A_i|}$ and # of simulator samples M chosen as $M_t = \lceil \frac{1}{\epsilon_t \gamma_t^2} \rceil$, the expected gap value of algorithm at time $T \geq 1$ satisfies:

$$\mathbb{E}[V(\pi_T)] \leq \frac{C_7}{T} + \frac{C_6 \log(T)}{T},$$

and further, $\lim_{T \rightarrow \infty} V(\pi(T)) = 0$ in probability.

QGRE Computation – Numerical Results

QRE for $\lambda_i \equiv 1$ and different games



GQRE Computation—Hard Instances

3-player game with 2 actions per player $\{a, b\}$ & utilities

$$\begin{bmatrix} (0, 0, 0) & (1, 1, 0) \\ (1, 0, 1) & (0, 1, 1) \end{bmatrix} \quad \begin{bmatrix} (0, 1, 1) & (1, 0, 1) \\ (1, 1, 0) & (0, 0, 0) \end{bmatrix}$$

Each player i wants to mismatch with $i + 1$ irrespective of $i - 1$

- Hart-MasColell2003: ANY dynamics using current gradient will not converge to NE for some games
- ToonsiShamma2023: Higher-order dynamics can converge
- Papadimitrou et al. 2023: No deterministic algorithm will converge for some games

QRE for Jordan1993's game

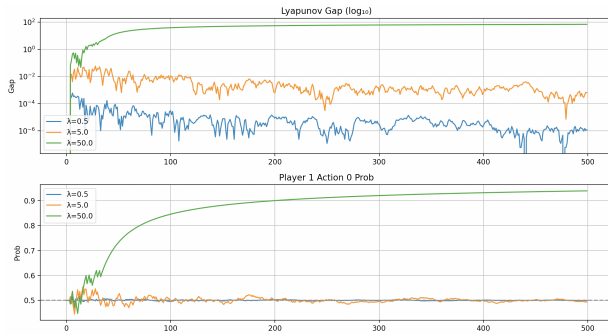


Figure: QRE computation for $\lambda_i = 0.5, 5, 50$

Negative definiteness of underlying utility matrix true when $\lambda \leq 1$

Agenda

- 1 Nash Equilibria (NE)
- 2 Generalized Quantal Response Equilibria
- 3 GQRE Computation with Learning
- 4 Conclusion

Conclusion

- Studied equilibrium notion based on perturbed utilities
- Fixed-point, existence and uniqueness characterization
- (Novel) Simulator oracle
- (Novel) Smoothened Frank-Wolfe algorithm & gap function
 - No-regret algorithm
- Hard game instances are nevertheless challenging
- Further weakening of NE tenets toward mixed agents?